

Ingest Logs into SIEM + Basic Searches

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1. Objective

The goal of this task was to get real data into the SIEM (Splunk) and practise writing search queries using Splunk's Search Processing Language (SPL): running a basic search, filtering for failed logins with a count, and applying a time-range filter.

2. Ingesting Sample Log Data

A sample authentication log (sample_auth.log) containing 20 login/logoff events — a mix of successful and failed logins across several hosts and IP addresses — was uploaded into Splunk via Add Data > Upload, with the source type configured to break the file into one event per line.

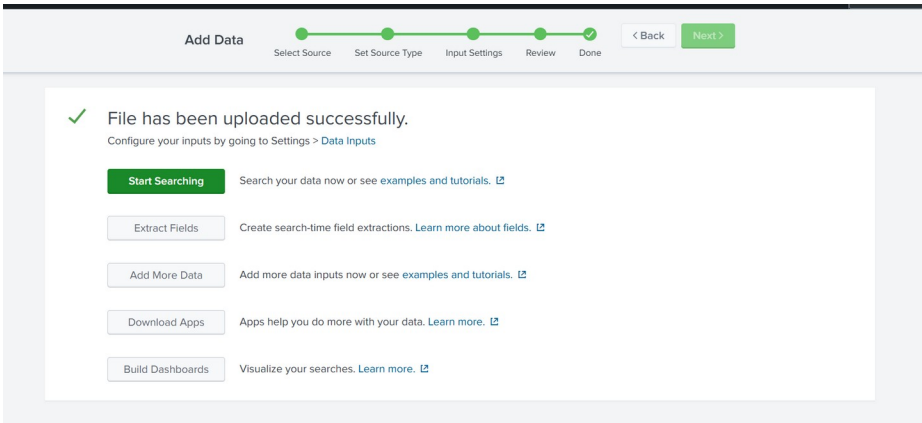


Figure 1: Confirmation that the sample log file was uploaded successfully into Splunk.

3. Searches Performed

Three SPL searches were run against the ingested data, building from a basic search to a filtered, counted, time-bound result, as required by the task.

#	Search Query (SPL)	Purpose	Result
1	source="sample_auth.log" host="Yusra-Imran" sourcetype="syslog"	Basic search — retrieve all ingested log data.	Ingested log data returned successfully.
2	source="sample_auth.log" "status=failed"	Filter the log data down to only failed login attempts.	All failed-logon lines isolated and highlighted.
3	source="sample_auth.log" head 1 eval failed_count=mvcount(split(_raw,"status=failed"))-1 table failed_count	Get an exact count of failed logins, with a 'Last 7 days' time-range filter applied.	failed_count = 12

3.1 Basic search

```
source="sample_auth.log" host="Yusra-Imran" sourcetype="syslog"
```

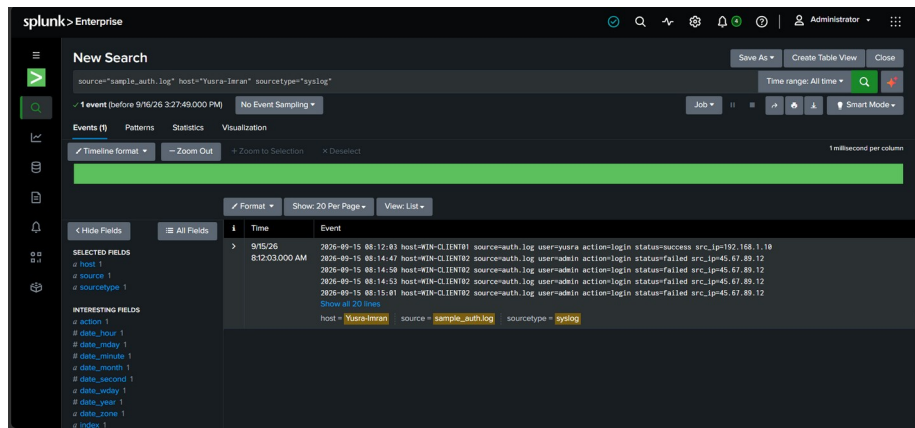


Figure 2: Basic search retrieving the ingested log data.

3.2 Filtering failed logins

```
source="sample_auth.log" "status=failed"
```

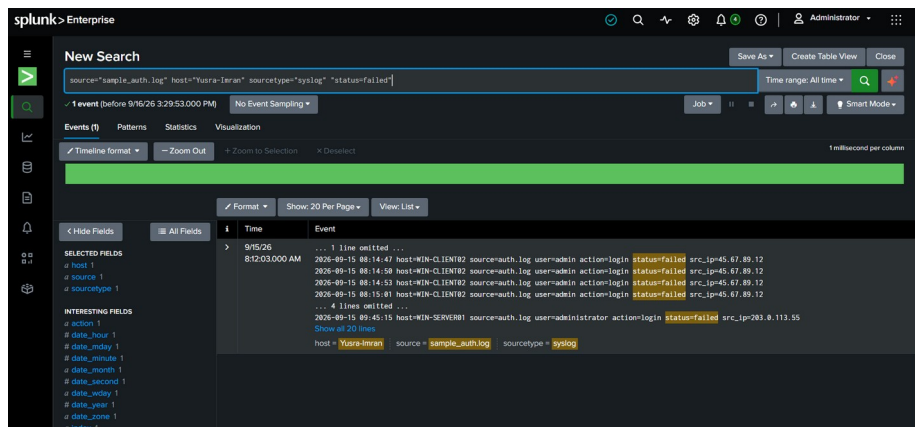


Figure 3: Search results filtered to only failed-login lines (highlighted in yellow).

3.3 Count with time-range filter

```
source="sample_auth.log" | head 1 | eval
failed_count=mvcount(split(_raw,"status=failed"))-1 | table failed_count
```

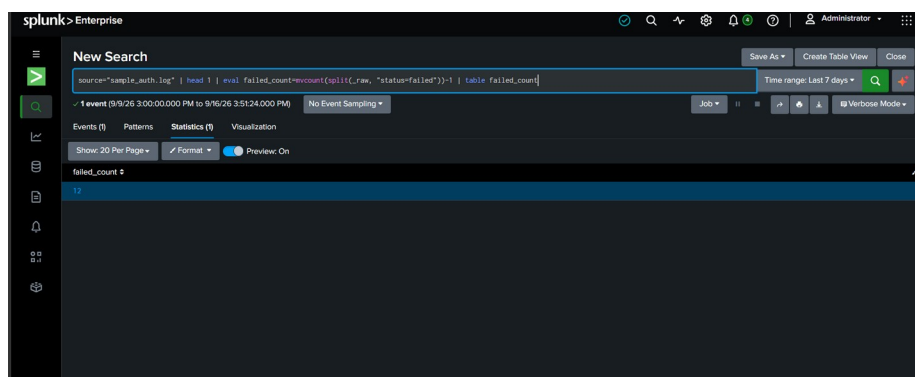


Figure 4: Exact count of failed logins (12) with the time range set to 'Last 7 days'.

4. What Was Found

Out of 20 total login-related events in the sample log, 12 were failed login attempts and 8 were successful logins/logoffs. Two accounts in particular — “admin” (from IP 45.67.89.12) and “administrator” (from IP 203.0.113.55) — accounted for

repeated, rapid failed attempts, a pattern consistent with brute-force login activity that a SOC analyst would flag for further investigation.

5. TryHackMe Room

The TryHackMe room listed as a resource for this task, “Introduction to SIEM,” was completed on the free tier (100%), as mandated by the task.

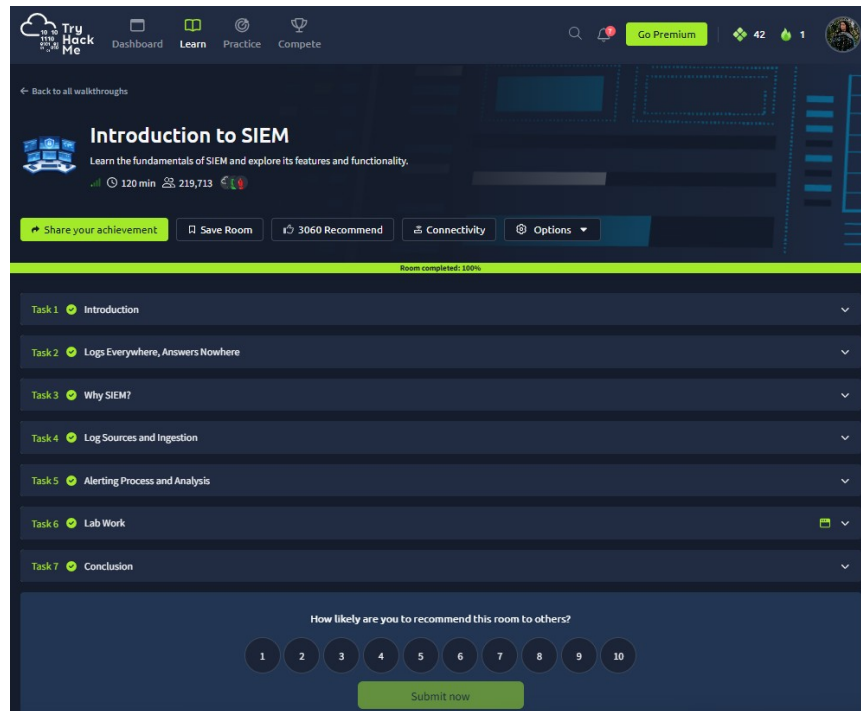


Figure 5: “Introduction to SIEM” room — Room completed: 100%.

Note: This room was originally completed as part of Task 10, where the same SIEM concept resource applies; it is shown again here since Task 11 also lists it as a required resource.

6. Conclusion

This task provided hands-on practice with the core SIEM workflow: getting log data into the platform, then using SPL to search, filter, count, and time-bound results. Isolating and counting failed logins — and spotting which accounts they targeted — demonstrates the basic investigative technique a SOC analyst uses every day inside a real SIEM.